# A Kernel-Checked Spatial Bridge for Vortex-Line Comparison: Lean 4 Formalization of the Deng–Hou–Yu Spatial Identity for Smooth Periodic Navier–Stokes Fields

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Repository: https://github.com/BenFrohman/Clay-

November 9, 2026 (America/Chicago, UTC−6); revised the same day for the smoothness-index correction (Section 2.3). Preprint; not peer reviewed.

**Abstract**

We describe a Lean 4/Mathlib formalization, merged in the public repository `BenFrohman/Clay-` at commit `43653aa`, of the *spatial* half of Theorem 1 of Deng, Hou and Yu (DHY): along a nonvanishing vortex-line segment from a point $x$ to a point $y$, the vorticity magnitudes satisfy $|\omega(x)| \le e^C |\omega(y)|$ whenever $\left|\int_0^L (\operatorname{div}\xi)(\gamma(s))\,ds\right| \le C$, where $\xi = \omega/|\omega|$. The library proves the named target `SpatialBridgeObligation` for every packet of solution-linked vortex-line data over every smooth periodic Navier–Stokes solution in the repository's encoding, with no assumed comparison, regularity supplement or integrability premise. The proof derives spatial $C^2$ regularity from the solution hypothesis, mixed-partial symmetry, $\operatorname{div}\operatorname{curl} = 0$, a product-rule identity for $|\omega|$ and $\xi$, a logarithmic chain rule along the curve, and the fundamental theorem of calculus. As a consequence the library proves that finiteness of the time integral of the endpoint vorticity implies finiteness of the Beale–Kato–Majda integral. All of the theorems are checked by the Lean kernel, and the full library axiom audit (133 theorems, 244 declarations after the correction below) reports only `propext`, `Classical.choice` and `Quot.sound`, with no `sorry`. We write every step of the argument out on paper. We also record three limitations, each of them kernel-checked: the argument uses only spatial $C^2$ smoothness, so it is kinematic and not specific to Navier–Stokes; the endpoint hypothesis is *equivalent* to Beale–Kato–Majda finiteness, so the criterion moves the analytic difficulty rather than removing it; and earlier versions of the repository's solution predicate and data class, written with a bare $\top$, required joint real-analyticity rather than $C^\infty$ smoothness at the pinned Mathlib. An informal, unformalized shear-flow argument shows that, granting the classical uniqueness of smooth solutions, that earlier formal version of Statement (B) was false as written. The repository now uses the $C^\infty$ index $\infty$; the full library still builds with the same proofs of the results here and the same axioms. **This is not a proof of the Clay Millennium Navier–Stokes problem, and it does not prove global regularity for any class of data.** Kernel verification certifies the Lean statements as written. It does not certify their fidelity to the intended PDE notions.

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# 1 Scope and non-claims

**What this paper is.** It is an exposition of a small, completely formalized piece of classical vorticity kinematics. It gives the exact Lean statements, file and line references at a pinned commit, a complete human-readable proof that follows the Lean proof step by step, and reproduction instructions. The mathematics is classical: it is the identity (3.1)–(3.2) of DHY Lemma 1 and the ratio bound (2.2) of DHY Theorem 1 [3]. *No novelty is claimed for the mathematics.* The contribution is the machine-checked formalization, its tie to the repository's definition of a Navier–Stokes solution, and a precise account of what remains open.

**What this paper is not.**

1. It is **not** a proof or a disproof of Fefferman's Statement (B) [4] or of any other Clay Millennium statement.

2. It does **not** prove that any nonzero Navier–Stokes solution admits a vortex-line packet (Definition 3.1). It also does not prove that a packet with time-uniform constants and measurable dependence on time exists.

3. It does **not** prove the endpoint integrability hypothesis (Definition 3.2) for any nonzero solution. Proposition 5.1 shows that this hypothesis is exactly as strong as finiteness of the Beale–Kato–Majda (BKM) integral.

4. It does **not** formalize the classical local existence theory or the BKM continuation theorem [1]. In the repository these appear only as the named, unproved proposition `PeriodicMaximalSolutionWithBKM`.

5. It does **not** formalize DHY Theorem 2, which is specific to the Euler equations, or the degenerate and general cases covered by DHY Theorem 1.

6. A successful kernel check means that the Lean terms typecheck against the Lean definitions. It does **not** show that those definitions faithfully encode Fefferman's notions of a smooth periodic solution, or any physical statement. Section 2.3 records a concrete fidelity issue of exactly this kind.

# 2 The formal setting

All declarations live in the repository [5] at commit `43653aab3b51269a71a713ccbc4d892869bbe497` on `main`. That commit is the merge of pull request #9, whose proof commit is `60e7ed0`. File and line numbers below refer to that commit; the later smoothness-index correction (Section 2.3) edits only the index on the same lines, so every cited *file:line* is unchanged. In references of the form *name (file:line)*, the file is relative to `lean/Clay/`; the pull request carrying this paper does not change any Lean library file. Lean is version 4.16.0 and Mathlib is tag `v4.16.0` (revision `a6276f4c6097675b1cf5ebd49b1146b735f38c02`) [2, 7].

## 2.1 Space, derivatives and vorticity

Let $E = \mathbb{R}^3$ with the Euclidean norm (`E3 := EuclideanSpace ℝ (Fin 3)`, `lean/Clay/StatementB.lean`:35), and let $e_1, e_2, e_3$ be the standard basis (`e` (`StatementB.lean`:38)). In Lean the indices are $0, 1, 2$; we write $1, 2, 3$. For $f\colon E \to \mathbb{R}$ the partial derivative is defined through the Fréchet derivative,

$$\partial_j f(x) := Df(x)[e_j]$$

(`pd` (`StatementB.lean`:41)). Mathlib's `fderiv` returns 0 at points where $f$ is not differentiable. Every use of $\partial_j$ below is at a point where differentiability has been proved, so this convention never enters. For $v\colon E \to E$,

$$\operatorname{curl} v = (\partial_2 v_3 - \partial_3 v_2,\ \partial_3 v_1 - \partial_1 v_3,\ \partial_1 v_2 - \partial_2 v_1)$$

(`curl` (`StatementB.lean`:53)), and $\operatorname{div} w = \sum_i \partial_i w_i$; the predicate $\operatorname{div} w \equiv 0$ is `DivFree` (`StatementB.lean`:50).

## 2.2 Solutions, lifespans and the BKM integral

For $T \in [0, \infty]$ the lifespan is $[0, T) = \{t : 0 \le t,\ \mathrm{ofReal}(t) < T\}$ (`lifespanSet` (`StatementB.lean`:82)). The predicate `NSOn` (`StatementB.lean`:65), written $\mathrm{NSOn}(\nu, S, u_0, u, p)$, asserts the following.

(N1) $(t, x) \mapsto u(t, x)$ and $(t, x) \mapsto p(t, x)$ satisfy `ContDiffOn ℝ ∞` on $S \times \mathbb{R}^3$ (jointly $C^\infty$; see Section 2.3).

(N2) The momentum equation holds with the one-sided time derivative `derivWithin` along $S$.

(N3) $\operatorname{div} u(t) = 0$ for $t \in S$.

(N4) $u(0, \cdot) = u_0$.

(N5) $u(t)$ and $p(t)$ are 1-periodic in each coordinate for $t \in S$.

The vorticity supremum is $\Omega(t) = \sup_x |\operatorname{curl} u(t)(x)| \in [0, \infty]$ (`vortSup` (`StatementB.lean`:95)), and

$$\mathrm{BKMIntegralFinite}(u, T) \ :\Longleftrightarrow\ \int_{[0,T)}^{-} \Omega(t)\, dt < \infty$$

(`BKMIntegralFinite` (`StatementB.lean`:98)), where $\int^-$ is the lower Lebesgue integral of a $[0,\infty]$-valued function. The quantifier `ForAllFiniteSolutions` (`StatementB.lean`:102) ranges over all $\nu > 0$, all data $u_0$ satisfying `SmoothPeriodicData`, all $(u,p)$, and all finite $T$ with $\mathrm{NSOn}(\nu,[0,T),u_0,u,p)$.

### 2.3 The smoothness index, and a correction

At the pinned Mathlib the smoothness index of `ContDiff` and `ContDiffOn` has type `WithTop` $\mathbb{N}\infty$. Mathlib writes $\infty$ for $((\top : \mathbb{N}_\infty) :$ `WithTop` $\mathbb{N}\infty)$, which means $C^\infty$, and $\omega$ for $\top :$ `WithTop` $\mathbb{N}\infty$, which means real-analytic (Mathlib file `Mathlib/Analysis/Calculus/ContDiff/FTaylorSeries.lean`, scoped notation `ContDiff`).

**Correction (November 9, 2026).** Versions of the repository up to and including commit `1849c26` wrote a bare $\top$ in `NSOn`, `SmoothPeriodicData`, and in the unasserted propositions `reversedHolderClaim` and `spikeObstruction`. That $\top$ is $\omega$, so those earlier formal statements, including `FeffermanStatementB`, `PeriodicMaximalSolutionWithBKM` and `PeriodicBKMIntegrability`, were statements about analytic data and analytic solutions, not Fefferman's $C^\infty$ statement. The repository now writes $\infty$ in all four places. After the change the full library builds; all 125 previously audited theorems check unchanged in name and axioms, and only the proof of `spatialC2_of_NSOn` needed a one-term edit (the bound $2 \le \infty$ instead of $2 \le \omega$). Theorems that take `NSOn` as a hypothesis now assume less and are therefore more general. The library module `Clay/Notes/SmoothnessIndex.lean` and the paper-check file `papers/spatial-bridge/PaperChecks.lean` kernel-check the following.

- $\infty \neq \omega$ in `WithTop` $\mathbb{N}\infty$.
- `NSOn_velocity_contDiffOn_infty` and `smoothPeriodicData_contDiff_infty`: the corrected predicates require $C^\infty$ (index $\infty$).
- `exists_contDiff_infty_not_omega`: the function `expNegInvGlue` ($e^{-1/x}$ for $x > 0$, 0 otherwise) is $C^\infty$ on $\mathbb{R}$ but not analytic at 0. So the earlier index was a strictly stronger requirement.
- `smoothPeriodicData_of_analytic`: analytic divergence-free periodic data remain admissible.

Fefferman's Statement (B) asks for $C^\infty$ data and $C^\infty$ solutions on $\mathbb{R}^3 \times [0,\infty)$, which is what the corrected `NSOn` and `SmoothPeriodicData` now say. A kernel check confirms only that the Lean terms typecheck against these definitions; it does not by itself certify that the definitions are a faithful encoding (Section 1). The following remark, kept for the record, explains why the earlier analytic version was not merely stronger but, granting classical uniqueness, false.

**Remark 2.1** (Informal, *not* formalized: a shear flow that is smooth but not time-analytic at $t = 0$)**.** Fix $\nu > 0$. Let

$$\varphi_0(r) = \sum_{k\in\mathbb{Z}} e^{-|k|} e^{2\pi i k r} = 1 + 2\sum_{k\ge 1} e^{-k}\cos(2\pi k r),$$

and put $u_0(x) = (0,\varphi_0(x_1),0)$. The series converges for $|\operatorname{Im} r| < 1/(2\pi)$, so $\varphi_0$ is real-analytic on all of $\mathbb{R}$, and $u_0$ is real-analytic, 1-periodic and divergence-free, since $\operatorname{div} u_0 = \partial_2\varphi_0(x_1) = 0$.

Let $\varphi(t,r) = \sum_k e^{-|k|-4\pi^2\nu k^2 t} e^{2\pi i k r}$ for $t \ge 0$, which solves $\varphi_t = \nu\varphi_{rr}$, and let $u(t,x) = (0,\varphi(t,x_1),0)$ with $p \equiv 0$. Then $(u\cdot\nabla)u = u_2\,\partial_2 u = 0$, because $u_1 = 0$ and $u$ does not depend on $x_2$, and $\Delta u = (0,\varphi_{rr},0)$. So $(u,p)$ is a smooth periodic solution of the unforced equations on $[0,\infty)\times\mathbb{R}^3$.

Now let $h(t) = \varphi(t,0)$. Every $t$-derivative series is dominated on $[0,\infty)$ by $\sum_k (4\pi^2\nu k^2)^n e^{-|k|} < \infty$, so the one-sided derivatives at 0 are

$$h^{(n)}(0) = \sum_k (-4\pi^2\nu k^2)^n e^{-|k|}.$$

All terms have the sign $(-1)^n$. Keeping only the term $k = 2n$ and using $n! \le n^n$,

$$\frac{|h^{(n)}(0)|}{n!} \;\ge\; \frac{(4\pi^2\nu)^n (2n)^{2n} e^{-2n}}{n^n} = \Big(\frac{16\pi^2\nu\, n}{e^2}\Big)^n.$$

The $n$-th root tends to $\infty$, so the Taylor series of $h$ at 0 has radius of convergence 0. If $(t,x) \mapsto u(t,x)$ were analytic within $[0,\infty)\times\mathbb{R}^3$ at $(0,0)$, then so would be its restriction $t \mapsto u_2(t,0) = h(t)$ within $[0,\infty)$. The coefficients of a power series that represents $h$ on $[0,r)$ are forced to be $h^{(n)}(0)/n!$, so that series would have positive radius. This is a contradiction. Hence this solution does *not* satisfy $\texttt{NSOn}(\nu,[0,\infty),u_0,u,p)$, although $u_0$ satisfies `SmoothPeriodicData`.

Smooth periodic solutions with given data are unique. This is classical, by the energy method, and it is *not* formalized. Assuming it, any `NSOn` solution for this $u_0$, which is in particular smooth, would coincide with the shear flow above, which is impossible. Granting classical uniqueness, the earlier (analytic) formal `FeffermanStatementB` was therefore *false*. The same datum made the earlier formal `PeriodicMaximalSolutionWithBKM` false as well, since that statement also requires `NSOn` on $[0,T)$ with $T > 0$, an interval containing $t = 0$. The reduction `feffermanStatementB_of_bkm` was and remains a correct implication. With the corrected index $\infty$ this obstruction disappears: the shear flow above is $C^\infty$ on $[0,\infty)\times\mathbb{R}^3$, so it is an `NSOn` solution for its datum. This remark is informal and is not formalized.

## 3 Solution-linked vortex-line packets

For a velocity $u$ and a time $t$ write $\omega(t,\cdot) = \operatorname{curl} u(t)$ and $N(t,\cdot) = |\omega(t,\cdot)|$. The normalized vorticity is

$$\xi(t,x) = N(t,x)^{-1}\omega(t,x)$$

(`direction` (`Depletion/SolutionLinkedVortexLines.lean`:24)), with the convention $0^{-1} = 0$, so that $\xi = 0$ where $\omega = 0$. If $\omega(t,x) \neq 0$ then $|\xi(t,x)| = 1$ (`norm_direction_of_nonzero` (`Depletion/SolutionLinkedVortexLines.lean`:28)). Its divergence is $\operatorname{div}\xi = \sum_i \partial_i \xi_i$ (`directionDivergence` (`Depletion/SolutionLinkedVortexLines.lean`:34)).

**Definition 3.1** (Packet; `Packet` (`Depletion/SolutionLinkedVortexLines.lean`:56))**.** A *packet* $P$ over $(u,T)$ consists of the following data:

- maps $x, y\colon \mathbb{R} \to E$ (the peak and endpoint) and a curve $\gamma\colon \mathbb{R}\times\mathbb{R} \to E$;
- a length $L\colon \mathbb{R} \to \mathbb{R}$;
- constants $K \in \mathbb{R}_{\ge 0}$ with $K \ge 1$ and $C \in \mathbb{R}_{\ge 0}$.

For every $t \in [0,T)$ the data must satisfy:

(P1) $L(t) \ge 0$, $\gamma(t,0) = x(t)$ and $\gamma(t,L(t)) = y(t)$;

(P2) the near-maximum condition $\Omega(t) \le K\, N(t,x(t))$, in $[0,\infty]$;

(P3) if $\omega(t,x(t)) \neq 0$, then:

  (a) $L(t) > 0$;

(b) $\omega(t, \gamma(t, s)) \neq 0$ for all $s \in [0, L(t)]$;

(c) $s \mapsto \gamma(t, s)$ has derivative $\xi(t, \gamma(t, s))$ within $[0, L(t)]$ at every $s \in [0, L(t)]$, so $\gamma(t, \cdot)$ is a unit-speed parametrization of a vortex line;

(d) $s \mapsto (\operatorname{div} \xi)(t, \gamma(t, s))$ is interval-integrable on $[0, L(t)]$;

(P4) the line control $\Lambda(t) := \left|\int_0^{L(t)} (\operatorname{div} \xi)(t, \gamma(t, s))\, ds\right|$ satisfies $\Lambda(t) \leq C$ (`lineControl`, line 46; set to 0 off $[0, T)$).

In addition the following maps must be Borel measurable on all of $\mathbb{R}$: $x$, $y$, $L$, $\Lambda$, and the endpoint magnitude

$$m(t) := \begin{cases} N(t, y(t)) & t \in [0, T), \\ 0 & \text{otherwise} \end{cases}$$

(`endpointMagnitude` (`Depletion/SolutionLinkedVortexLines.lean`:39)).

Every quantity in (P2)–(P4) and in $m$ is evaluated from the actual field $\operatorname{curl} u(t)$. There is no free magnitude or direction field that could be filled in independently of $u$. This is the difference from the legacy free interface `DHYPacket` (`lean/Clay/Gaps/DirectionAlignment.lean`:28), which is preserved unchanged. The constant $K$ is the reciprocal of DHY's near-maximum fraction. Condition (P3a) is an *extra* restriction of this implementation: DHY do not exclude degenerate segments, and we do not claim that they do.

**Definition 3.2** (Endpoint hypothesis and spatial comparison)**.** For a packet $P$:

- `EndpointIntegralFinite` (`Depletion/SolutionLinkedVortexLines.lean`:87) is the statement $\int^-_{[0,T)} m(t)\, dt < \infty$;
- `SpatialComparison` (`Depletion/SolutionLinkedVortexLines.lean`:92) is the statement $N(t, x(t)) \leq e^C\, m(t)$ in $[0, \infty]$ for all $t \in [0, T)$;
- `SpatialBridgeObligation` (`Depletion/SolutionLinkedVortexLines.lean`:101) is the statement that `SpatialComparison` holds for every packet over every solution in `ForAllFiniteSolutions`.

**Remark 3.3** (Nonvacuity, and its limit)**.** The zero field is a solution (`zero_NSOn`, line 204). It carries the packet `zeroPacket` (line 164), which satisfies the comparison, the endpoint hypothesis and BKM finiteness (`zeroPacket_BKMIntegralFinite`, line 229). Conversely, if $L(t) = 0$ then $\operatorname{curl} u(t) \equiv 0$ everywhere (`curl_eq_zero_of_length_eq_zero` (`Depletion/SolutionLinkedVortexLines.lean`:117)), so a zero-length placeholder cannot certify a field with nonzero vorticity. The library contains *no* packet over a nonzero solution. Nonvacuity is therefore established only for the zero solution.

# 4 The spatial bridge: statements and complete proofs

The main results are the following two theorems.

**Theorem 4.1** (`spatialBridgeObligation_proved` (`Depletion/ActualSpatialBridge.lean`:173))**.** `SpatialBridgeObligation` *holds. More generally (*`spatialComparison_of_NSOn` (`Depletion/ActualSpatialBridge.lean`:166)*), if* $\mathrm{NSOn}(\nu, [0, T), u_0, u, p)$ *holds for some* $T \in [0, \infty]$*, which may be infinite, then every packet* $P$ *over* $(u, T)$ *satisfies, for all* $t \in [0, T)$*,*

$$|\omega(t, x(t))| \;\leq\; e^C\, |\omega(t, y(t))|.$$

**Theorem 4.2** (`bkmIntegralFinite_of_NSOn_and_endpoint` (`Depletion/ActualSpatialBridge.lean`:179)). *If* $\mathrm{NSOn}(\nu, [0,T), u_0, u, p)$ *holds and* $P$ *is a packet over* $(u,T)$ *satisfying* `EndpointIntegralFinite`*, then* $\mathrm{BKMIntegralFinite}(u,T)$ *holds.*

The Lean statements, verbatim from the repository, are:

```
theorem spatialComparison_of_NSOn
    {ν : ℝ} {u0 : E3 → E3} {u : ℝ → E3 → E3} {p : ℝ → E3 → ℝ} {T : ℝ≥0∞}
    (hsol : NSOn ν (lifespanSet T) u0 u p) (P : Packet u T) : SpatialComparison P

theorem spatialBridgeObligation_proved : SpatialBridgeObligation

theorem bkmIntegralFinite_of_NSOn_and_endpoint
    {ν : ℝ} {u0 : E3 → E3} {u : ℝ → E3 → E3} {p : ℝ → E3 → ℝ} {T : ℝ≥0∞}
    (hsol : NSOn ν (lifespanSet T) u0 u p) (P : Packet u T)
    (hend : EndpointIntegralFinite P) : BKMIntegralFinite u T
```

The proof is a chain of seven lemmas, each a named Lean theorem. We give every step.

## 4.1 Step 1: spatial $C^2$ regularity from the solution hypothesis

**Lemma 4.3** (`spatialC2_of_NSOn` (`Depletion/ActualSpatialBridge.lean`:60)). *If* $\mathrm{NSOn}(\nu, S, u_0, u, p)$ *and* $t \in S$*, then each component* $y \mapsto u(t,y)_a$ *is* $C^2$ *on* $E$*.*

*Proof.* Let $\iota_t \colon E \to \mathbb{R} \times E$, $\iota_t(y) = (t,y)$. This map is the product of a constant map and the identity, so it is $C^\infty$; it maps $E$ into $S \times E$ because $t \in S$. By (N1), $U(z) = u(z_1, z_2)$ is $C^\infty$ on $S \times E$. The composition rule for `ContDiffOn` then shows that $U \circ \iota_t = u(t,\cdot)$ is $C^\infty$ on univ, that is, $C^\infty$ on $E$. A map into $E =$ `EuclideanSpace` $\mathbb{R}$ `(Fin 3)` is $C^n$ if and only if each coordinate is (`contDiff_piLp`). Finally $2 \le \infty$ lowers the index (`ContDiff.of_le`). The case $t = 0$ needs no separate treatment, because the composition is taken within $S \times E$. □

## 4.2 Step 2: symmetry of mixed partials

**Lemma 4.4** (`pd_pd_comm` (`Depletion/ActualSpatialBridge.lean`:11)). *If* $f \colon E \to \mathbb{R}$ *is* $C^2$*, then* $\partial_i(\partial_j f)(x) = \partial_j(\partial_i f)(x)$ *for all* $i, j, x$*.*

*Proof.* Since $f$ is $C^2$, $Df$ is differentiable at $x$. For a fixed vector $a$, the map $y \mapsto Df(y)[a]$ is the evaluation of the differentiable map $Df$ at a constant, so its derivative at $x$ in direction $b$ is $D^2 f(x)[b][a]$ (`fderiv_clm_apply`). Hence $\partial_i(\partial_j f)(x) = D^2 f(x)[e_i][e_j]$ and $\partial_j(\partial_i f)(x) = D^2 f(x)[e_j][e_i]$. Schwarz's theorem for $C^2$ maps (`ContDiffAt.isSymmSndFDerivAt`) gives $D^2 f(x)[e_i][e_j] = D^2 f(x)[e_j][e_i]$. □

## 4.3 Step 3: the divergence of a curl vanishes

**Lemma 4.5** (`div_curl_eq_zero` (`Depletion/ActualSpatialBridge.lean`:24)). *If every component of* $v \colon E \to E$ *is* $C^2$*, then* $\operatorname{div}\operatorname{curl} v \equiv 0$*.*

*Proof.* Fix $x$. Each $\partial_j v_a$ is differentiable at $x$, because $v_a$ is $C^2$. The derivative is linear, so for differentiable $f, g$ we have $\partial_i(f-g)(x) = \partial_i f(x) - \partial_i g(x)$. Expanding the definition of curl gives

$$\begin{aligned}\operatorname{div}\operatorname{curl} v(x) &= \partial_1(\partial_2 v_3 - \partial_3 v_2) + \partial_2(\partial_3 v_1 - \partial_1 v_3) + \partial_3(\partial_1 v_2 - \partial_2 v_1)\\ &= \partial_1\partial_2 v_3 - \partial_1\partial_3 v_2 + \partial_2\partial_3 v_1 - \partial_2\partial_1 v_3 + \partial_3\partial_1 v_2 - \partial_3\partial_2 v_1.\end{aligned}$$

Lemma 4.4 applied to $v_3$, $v_2$ and $v_1$ gives $\partial_1\partial_2 v_3 = \partial_2\partial_1 v_3$, $\partial_1\partial_3 v_2 = \partial_3\partial_1 v_2$ and $\partial_2\partial_3 v_1 = \partial_3\partial_2 v_1$. Substituting these and grouping terms,

$$(\partial_2\partial_1 v_3 - \partial_2\partial_1 v_3) + (\partial_3\partial_1 v_2 - \partial_3\partial_1 v_2) + (\partial_3\partial_2 v_1 - \partial_3\partial_2 v_1) = 0.$$

In Lean the three substitutions are `pd_pd_comm` at $(0,1)$, $(0,2)$ and $(1,2)$, and the final cancellation is `ring`. Neither the momentum equation nor $\operatorname{div} u = 0$ is used. □

**Lemma 4.6** (`curl_differentiable_of_C2` (`Depletion/ActualSpatialBridge.lean`:45)). *Under the hypothesis of Lemma 4.5,* $\operatorname{curl} v$ *is differentiable on* $E$.

*Proof.* A map into $E$ is differentiable at $x$ if and only if each coordinate is (`differentiableAt_piLp`). Each coordinate of $\operatorname{curl} v$ is a difference of two first partials $\partial_j v_a$, each differentiable at $x$ as in the previous proof. □

### 4.4 Step 4: the product-rule identity for $|\omega|$ and $\xi$

**Lemma 4.7** (`norm_direction_derivative` (`Depletion/ActualSpatialBridge.lean`:72)). *Let* $w = \operatorname{curl} v$. *Suppose that* $w$ *is differentiable at* $x$, *that* $w(x) \neq 0$, *and that* $\sum_i \partial_i w_i(x) = 0$. *Put* $N = |w|$ *and* $\xi = N^{-1} w$. *Then*

$$DN(x)[\xi(x)] \;=\; -N(x)\,(\operatorname{div}\xi)(x).$$

*Proof. Differentiability.* The Euclidean norm is differentiable away from 0, and $w(x) \neq 0$, so $N$ is differentiable at $x$ (`DifferentiableAt.norm`). Since $N(x) \neq 0$, $N^{-1}$ is differentiable at $x$, and hence so is the product $\xi = N^{-1} w$, together with each coordinate $\xi_i$.

*A global factorization.* For every $y \in E$,

$$w_i(y) = N(y)\,\xi_i(y).$$

If $w(y) = 0$, both sides are 0, because $\xi(y) = 0^{-1} \cdot 0 = 0$. If $w(y) \neq 0$, then $N(y)\xi_i(y) = N(y)N(y)^{-1} w_i(y) = w_i(y)$. The factorization holds as an equality of *functions*, which is what allows us to differentiate it.

*Product rule.* Differentiating at $x$ in the direction $e_i$ gives

$$\partial_i w_i(x) = N(x)\,\partial_i \xi_i(x) + \xi_i(x)\,\partial_i N(x).$$

*Summation.* Summing over $i = 1, 2, 3$ and using the hypothesis $\sum_i \partial_i w_i(x) = 0$:

$$0 = N(x) \sum_i \partial_i \xi_i(x) + \sum_i \xi_i(x)\, DN(x)[e_i] = N(x)\,(\operatorname{div}\xi)(x) + DN(x)\Big[\sum_i \xi_i(x) e_i\Big].$$

The last equality uses linearity of $DN(x)$. Since $\xi(x) = \sum_i \xi_i(x) e_i$, this is $0 = N(x)(\operatorname{div}\xi)(x) + DN(x)[\xi(x)]$, which rearranges to the claim. □

This is exactly DHY (3.1). Their computation is $0 = \nabla \cdot (|\omega|\xi) = (\xi \cdot \nabla)|\omega| + (\nabla \cdot \xi)|\omega|$, and the term $DN(x)[\xi(x)]$ is the directional derivative $(\xi \cdot \nabla)|\omega|$.

### 4.5 Step 5: the logarithmic derivative along the vortex line

**Lemma 4.8** (`logDerivativeAlong_of_spatialC2` (`Depletion/ActualSpatialBridge.lean`:115)). *Let* $P$ *be a packet over* $(u, T)$, *and suppose that every* $u(t)$ *with* $t \in [0, T)$ *has* $C^2$ *components. Let* $t \in [0, T)$ *with* $\omega(t, x(t)) \neq 0$, *and put* $L = L(t)$, $\gamma = \gamma(t, \cdot)$. *Then* $f(s) := \log N(t, \gamma(s))$ *has derivative* $-(\operatorname{div}\xi)(t, \gamma(s))$ *within* $[0, L]$ *at every* $s \in [0, L]$. *This is the predicate* `LogDerivativeAlong` *(line 107).*

*Proof.* Fix $s \in [0, L]$ and write $z = \gamma(s)$. By (P3b), $\omega(t, z) \neq 0$. By Lemma 4.6, $\omega(t, \cdot)$ is differentiable at $z$, so $N(t, \cdot)$ is differentiable at $z$, as in the proof of Lemma 4.7. By (P3c), $\gamma$ has derivative $\xi(t, z)$ within $[0, L]$ at $s$. The chain rule for a Fréchet-differentiable outer map composed with a within-differentiable curve (`HasFDerivAt.comp_hasDerivWithinAt`) shows

that $s \mapsto N(t, \gamma(s))$ has derivative $DN(t,\cdot)(z)[\xi(t,z)]$ within $[0, L]$ at $s$. By Lemma 4.5, $\operatorname{div}\omega(t,\cdot) = 0$ at $z$. All hypotheses of Lemma 4.7 therefore hold at $z$, and the derivative equals $-N(t,z)(\operatorname{div}\xi)(t,z)$. Since $N(t,z) > 0$, the chain rule for log (`HasDerivWithinAt.log`) gives

$$f'(s) = \frac{-N(t,z)\,(\operatorname{div}\xi)(t,z)}{N(t,z)} = -(\operatorname{div}\xi)(t,z)$$

within $[0, L]$. The division is legitimate because $N(t,z) \neq 0$; in Lean this step is `field_simp` followed by `ring`. □

### 4.6 Step 6: integration and exponentiation

**Lemma 4.9** (`spatialComparison_of_logDerivativeAlong` (`Depletion/ActualSpatialBridge.lean`:135)). *If $P$ satisfies* `LogDerivativeAlong`*, then* `SpatialComparison`$(P)$ *holds.*

*Proof.* Fix $t \in [0, T)$ and write $N_x = N(t, x(t))$, $N_y = N(t, y(t))$ and $L = L(t)$.

*Case $N_x = 0$.* The left side of the comparison is 0, and $0 \leq$ anything in $[0, \infty]$.

*Case $N_x \neq 0$.* Then $\omega(t, x(t)) \neq 0$, so (P3a) gives $L > 0$. By (P3b) at $s = L$ and (P1), $\omega(t, y(t)) = \omega(t, \gamma(t, L)) \neq 0$, so $N_y > 0$ (`endpoint_nonzero_of_peak_nonzero`, line 127). Let $f(s) = \log N(t, \gamma(t,s))$ and $g(s) = (\operatorname{div}\xi)(t, \gamma(t,s))$. We check the hypotheses of the fundamental theorem of calculus in the form `intervalIntegral.integral_eq_sub_of_hasDerivAt_of_le`:

- $0 \leq L$;
- $f$ is continuous on $[0, L]$, because a function with a derivative within a set at a point is continuous within that set at that point;
- at each interior point $s \in (0, L)$, the set $[0, L]$ is a neighbourhood of $s$, so the within-derivative $-g(s)$ is a genuine derivative;
- $-g$ is interval-integrable on $[0, L]$ by (P3d).

Therefore

$$\int_0^L (-g(s))\,ds = f(L) - f(0) = \log N_y - \log N_x,$$

using $\gamma(t,0) = x(t)$ and $\gamma(t,L) = y(t)$ from (P1). With $I := \int_0^L g(s)\,ds$ and $\int_0^L (-g) = -I$, this rearranges to

$$\log N_x = \log N_y + I.$$

By (P4), $I \leq |I| = \Lambda(t) \leq C$, hence $\log N_x \leq C + \log N_y$. Since exp is monotone, $\exp(\log a) = a$ for $a > 0$, and $\exp(C + \log N_y) = e^C N_y$, we obtain

$$N_x \leq e^C N_y.$$

Finally we pass to $[0, \infty]$. Since $N_x$ and $N_y$ are nonnegative, the coercions satisfy $\|\cdot\|_+ = \mathrm{ofReal}\|\cdot\|$. `ENNReal.ofReal` is monotone, and $\mathrm{ofReal}(e^C N_y) = \mathrm{ofReal}(e^C)\cdot\mathrm{ofReal}(N_y)$ because $e^C \geq 0$. Moreover $m(t) = N_y$ for $t \in [0, T)$. This gives the inequality required by `SpatialComparison`. □

Only the upper bound $N_x \leq e^C N_y$ is formalized. The reverse inequality $N_y \leq e^C N_x$ of DHY (2.2) follows from the same identity, since $\log N_y = \log N_x - I \leq \log N_x + C$, but it is not stated in the library.

### 4.7 Step 7: assembly

*Proof of Theorem 4.1.* By Lemma 4.3 with $S = [0,T)$, every slice $u(t)$ with $t \in [0,T)$ has $C^2$ components. Lemma 4.8 then gives `LogDerivativeAlong`$(P)$, and Lemma 4.9 gives `SpatialComparison`$(P)$. This proves `spatialComparison_of_NSOn`. For `spatialBridgeObligation_proved`, we unfold `ForAllFiniteSolutions`, introduce $\nu$, $\nu > 0$, $u_0, u, p, T$, the data hypothesis, the hypothesis $T \neq \top$ and $h_{\mathrm{sol}}$, and apply the previous statement. The hypotheses $\nu > 0$, the data hypothesis and $T \neq \top$ are not used. □

*Proof of Theorem 4.2.* This is `bkmIntegralFinite_of_spatialComparison_and_endpoint` (`Depletion/SolutionLinkedVortexLines.lean`:135) applied to the comparison from Theorem 4.1. Put $K' = K \cdot \mathrm{ofReal}(e^C)$. Then $K' < \infty$, because $K \in \mathbb{R}_{\geq 0}$ and $e^C < \infty$. For $t \in [0,T)$, (P2) and the comparison give

$$\Omega(t) \leq K\, N(t, x(t)) \leq K\, e^C m(t) = K' m(t).$$

The function $K'm$ is measurable, because $m$ is measurable by the packet's measurability field. The monotonicity of the set lower integral, `setLIntegral_mono`, requires measurability only of the majorant, and it gives $\int^-_{[0,T)} \Omega \leq \int^-_{[0,T)} K'm$. Measurability of $m$ also lets the constant come out of the integral (`lintegral_const_mul`): $\int^-_{[0,T)} K'm = K' \int^-_{[0,T)} m$. The right side is a product of two finite elements of $[0,\infty]$, hence finite. □

**Corollary 4.10** (only spatial $C^2$ is used; `spatialComparison_of_spatialC2` in `PaperChecks.lean`)**.** *Let $u\colon \mathbb{R} \to E \to E$ be any family whose slices $u(t)$, $t \in [0,T)$, have $C^2$ components. Then every packet over $(u,T)$ satisfies* `SpatialComparison`*. No evolution equation, viscosity, incompressibility of $u$, periodicity or time regularity is needed.*

*Proof.* Compose Lemma 4.8 with Lemma 4.9. The Lean term is `spatialComparison_of_logDerivativeAlongP(logDerivativeAlong_of_spatialC2PhC2)`. □

**Remark 4.11.** Corollary 4.10 makes precise that the "Navier–Stokes" content of Theorem 4.1 is only that solutions are spatially $C^2$. The statement is kinematic: it relates vorticity magnitudes *at one time* along one vortex line. This agrees with DHY, whose Theorem 1 is stated for "any 3-D incompressible flow (Euler or Navier–Stokes)", and whose Lemma 1 needs only $\omega(\cdot,t) \in C^1$.

## 5 What the BKM consequence does and does not give

**Proposition 5.1** (`endpointIntegralFinite_iff_bkm` in `PaperChecks.lean`)**.** *For every packet $P$ over $(u,T)$,* BKMIntegralFinite$(u,T)$ *implies* `EndpointIntegralFinite`$(P)$*. If in addition* NSOn$(\nu,[0,T),u_0,u,p)$ *holds, then the two statements are equivalent.*

*Proof.* For $t \in [0,T)$ we have $m(t) = N(t,y(t)) \leq \sup_z N(t,z) = \Omega(t)$, and for $t \notin [0,T)$ we have $m(t) = 0 \leq \Omega(t)$. So $m \leq \Omega$ pointwise on all of $\mathbb{R}$. Monotonicity of the lower Lebesgue integral for the restricted measure (`lintegral_mono`) needs no measurability, and it gives $\int^-_{[0,T)} m \leq \int^-_{[0,T)} \Omega < \infty$. The converse direction is Theorem 4.2. □

Proposition 5.1 is the honest summary of the analytic content. *Given* a packet, the vortex-line comparison converts the Clay-core quantity $\int \Omega$ into the endpoint quantity $\int m$, and back, with no loss. The criterion helps only when one has independent control of the vorticity at the far endpoint $y(t)$, as in the numerical scenarios DHY discuss, where $y(t)$ lies outside a collapsing tube. If $y(t)$ is chosen near the peak, the endpoint hypothesis is no easier than BKM itself. In the repository's terms, the Clay core `PeriodicBKMIntegrability` (`lean/Clay/StatementB.lean`:117) remains *open*. Nothing here reduces it to a weaker or a checkable hypothesis for general data.

# 6 Ledger: proved, assumed, and open

### Kernel-checked in the library (commit `43653aa`, rechecked after the smoothness-index correction)

The following are proved in Lean with standard axioms only:

- the ten named theorems of `lean/Clay/Depletion/ActualSpatialBridge.lean` (Table 1);
- the supporting theorems of `SolutionLinkedVortexLines.lean` listed in the table;
- the conditional reduction `feffermanStatementB_of_bkm` (`lean/Clay/StatementB.lean`:122), from the earlier monograph [6];
- the smoothness-index checks of `lean/Clay/Notes/SmoothnessIndex.lean` (Section 2.3).

### Kernel-checked in this paper's check file (not part of the library audit)

- `NSOn_velocity_contDiffOn_infty`, `smoothPeriodicData_contDiff_infty` and `contDiff_infty_ne_omega` (Section 2.3);
- `spatialComparison_of_spatialC2` (Corollary 4.10);
- `endpointIntegralFinite_of_bkm` and `endpointIntegralFinite_iff_bkm` (Proposition 5.1).

The pull request's CI runs this file and prints `#printaxioms` for each declaration.

### Hypotheses carried by the theorems (assumed, not proved)

1. **All of `NSOn`.** This means (N1)–(N5), including joint $C^\infty$ regularity (Section 2.3). Only the spatial $C^2$ consequence of (N1) is used.
2. **All packet fields (P1)–(P4)** and the five measurability fields, for the specific packet supplied: near-maximality with a fixed $K$, nonvanishing along the segment, the unit-speed vortex-line tangent equation, integrability of $\operatorname{div}\xi$ along the segment, and the time-uniform bound $C$.
3. **`EndpointIntegralFinite`** (Theorem 4.2 only), a separate argument which by Proposition 5.1 is as strong as BKM finiteness.

### Classical inputs *not* formalized

1. Local existence and uniqueness of smooth periodic solutions, and the BKM continuation criterion [1]. These appear only as the named proposition `PeriodicMaximalSolutionWithBKM` (`lean/Clay/StatementB.lean`:109). The step from BKM finiteness to "no blow-up" therefore has no formal proof here. In particular Theorem 4.2 concludes integral finiteness, *not* regularity.
2. Existence of vortex-line packets for nonzero solutions: integral curves of $\xi$ through $x(t)$, their nonvanishing extent, a measurable choice of $x(t), y(t), L(t)$ in $t$, and time-uniform constants $K$ and $C$. None of this is proved for any nonzero solution.
3. The general and degenerate configurations of DHY Theorem 1 (for example $L(t) = 0$ with nonzero vorticity), and the two-sided ratio bound (2.2).

4. DHY Theorem 2 and Lemmas 2–3. These use the Euler vorticity transport $\omega(X(\alpha,t),t) = \nabla_\alpha X\, \omega(\alpha,t_1)$, which fails for Navier–Stokes. None of them is imported.

### Open

The Clay core `PeriodicBKMIntegrability` and Fefferman's Statement (B) remain open. The fidelity issue of Section 2.3 is corrected in the library: the formal `FeffermanStatementB` now asks for $C^\infty$ solutions of $C^\infty$ data. Earlier versions required analytic solutions and, by Remark 2.1 (informal, and conditional on classical uniqueness), were false as written.

## 7 Declaration index

Locations are given relative to `lean/Clay/` at commit `43653aab3b51269a71a713ccbc4d892869bbe497`. The CI script `papers/spatial-bridge/check_refs.py` checks every *file:line* pair printed in this paper: it confirms that the cited line declares the cited name.

Table 1: Lean declarations used in this paper.

| Declaration | Location |
|---|---|
| *New in pull request #9 (*`ActualSpatialBridge.lean`*); ten named theorems* | |
| `pd_pd_comm` (`Depletion/ActualSpatialBridge.lean`:11) | Lemma 4.4 |
| `div_curl_eq_zero` (`Depletion/ActualSpatialBridge.lean`:24) | Lemma 4.5 |
| `curl_differentiable_of_C2` (`Depletion/ActualSpatialBridge.lean`:45) | Lemma 4.6 |
| `spatialC2_of_NSOn` (`Depletion/ActualSpatialBridge.lean`:60) | Lemma 4.3 |
| `norm_direction_derivative` (`Depletion/ActualSpatialBridge.lean`:72) | Lemma 4.7 |
| `LogDerivativeAlong` (`Depletion/ActualSpatialBridge.lean`:107) | definition (Prop), Lemma 4.8 |
| `logDerivativeAlong_of_spatialC2` (`Depletion/ActualSpatialBridge.lean`:115) | Lemma 4.8 |
| `spatialComparison_of_logDerivativeAlong` (`Depletion/ActualSpatialBridge.lean`:135) | Lemma 4.9 |
| `spatialComparison_of_NSOn` (`Depletion/ActualSpatialBridge.lean`:166) | Theorem 4.1 |
| `spatialBridgeObligation_proved` (`Depletion/ActualSpatialBridge.lean`:173) | Theorem 4.1 |
| `bkmIntegralFinite_of_NSOn_and_endpoint` (`Depletion/ActualSpatialBridge.lean`:179) | Theorem 4.2 |
| *From pull request #8 (*`SolutionLinkedVortexLines.lean`*)* | |
| `direction` (`Depletion/SolutionLinkedVortexLines.lean`:24) | $\xi$ |
| `norm_direction_of_nonzero` (`Depletion/SolutionLinkedVortexLines.lean`:28) | $\lvert\xi\rvert = 1$ |
| `directionDivergence` (`Depletion/SolutionLinkedVortexLines.lean`:34) | $\operatorname{div}\xi$ |
| `endpointMagnitude` (`Depletion/SolutionLinkedVortexLines.lean`:39) | $m$ |
| `lineControl` (`Depletion/SolutionLinkedVortexLines.lean`:46) | $\Lambda$ |

| Declaration | Location |
|---|---|
| `Packet` (`Depletion/SolutionLinkedVortexLines.lean`:56) | Definition 3.1 |
| `EndpointIntegralFinite` (`Depletion/SolutionLinkedVortexLines.lean`:87) | Definition 3.2 |
| `SpatialComparison` (`Depletion/SolutionLinkedVortexLines.lean`:92) | Definition 3.2 |
| `SpatialBridgeObligation` (`Depletion/SolutionLinkedVortexLines.lean`:101) | Definition 3.2 |
| `curl_eq_zero_of_length_eq_zero` (`Depletion/SolutionLinkedVortexLines.lean`:117) | zero-length rejection |
| `endpoint_nonzero_of_peak_nonzero` (`Depletion/SolutionLinkedVortexLines.lean`:127) | Lemma 4.9 |
| `bkmIntegralFinite_of_spatialComparison_and_endpoint` (`Depletion/SolutionLinkedVortexLines.lean`:135) | Theorem 4.2 |
| `zeroPacket` (`Depletion/SolutionLinkedVortexLines.lean`:164) | nonvacuity (zero field) |
| `zero_NSOn` (`Depletion/SolutionLinkedVortexLines.lean`:204) | nonvacuity (zero field) |
| `zeroPacket_BKMIntegralFinite` (`Depletion/SolutionLinkedVortexLines.lean`:229) | nonvacuity (zero field) |
| *Base definitions (*`StatementB.lean`*)* | |
| `pd` (`StatementB.lean`:41) | $\partial_j$ |
| `DivFree` (`StatementB.lean`:50) | $\operatorname{div} = 0$ |
| `curl` (`StatementB.lean`:53) | curl |
| `SmoothPeriodicData` (`StatementB.lean`:60) | data class (Section 2.3) |
| `NSOn` (`StatementB.lean`:65) | solution predicate |
| `FeffermanStatementB` (`StatementB.lean`:77) | open target |
| `lifespanSet` (`StatementB.lean`:82) | $[0, T)$ |
| `vortSup` (`StatementB.lean`:95) | $\Omega$ |
| `BKMIntegralFinite` (`StatementB.lean`:98) | BKM integral |
| `ForAllFiniteSolutions` (`StatementB.lean`:102) | quantifier |
| `PeriodicMaximalSolutionWithBKM` (`StatementB.lean`:109) | classical, *not formalized* |
| `PeriodicBKMIntegrability` (`StatementB.lean`:117) | Clay core, *open* |
| `feffermanStatementB_of_bkm` (`StatementB.lean`:122) | proved reduction |
| `DHYPacket` (`Gaps/DirectionAlignment.lean`:28) | legacy free interface |

**Axiom audit.** CI for the merge commit `43653aa` ran the library audit `lean/AxiomReport.lean`, which iterates over every non-internal constant in the `Clay` module tree and collects its axioms. It reported `AXSUMMARY 125 theorems, 236 declarations in Clay; nonstandard: 0`, and every row lists a subset of `propext`, `Classical.choice` and `Quot.sound`. After the smoothness-index correction a full local build and the same audit report `AXSUMMARY 133 theorems, 244 declarations in Clay; nonstandard: 0`; the 125 earlier rows are unchanged and the 8 new rows belong to `Clay.Notes.SmoothnessIndex`. The workflow fails on any occurrence of `sorryAx` and on any nonstandard axiom. The count of 125 includes compiler-generated equation and structure lemmas. Twelve of the 125 rows are attributed to the new module: the ten named theorems above, the auxiliary `LogDerivativeAlong.proof_1`, and the equation lemma `Clay.StatementB.e.eq_1`, which Lean generates on demand in that

module.

## 8 Reproducibility

- **Repository:** https://github.com/BenFrohman/Clay-.
- **Library commit:** `43653aab3b51269a71a713ccbc4d892869bbe497` (`main`; merge of pull request #9). The Lean files cited here are unchanged on the paper branch, except for the smoothness-index correction of Section 2.3.
- **Toolchain:** `leanprover/lean4:v4.16.0` (`lean/lean-toolchain`).
- **Mathlib:** `v4.16.0`, revision `a6276f4c6097675b1cf5ebd49b1146b735f38c02`, pinned in `lean/lake-manifest.json` together with its transitive dependencies.
- **CI:** the workflow `.github/workflows/lean.yml` (Lean build, axiom audit, regressions and paper checks) and `.github/workflows/paper-spatial-bridge.yml` (reference check and LaTeX build).

To reproduce, run:

```
git clone https://github.com/BenFrohman/Clay- && cd Clay-
git checkout 43653aab3b51269a71a713ccbc4d892869bbe497   # or the paper branch
cd lean
lake exe cache get          # prebuilt Mathlib for the pinned revision
lake build                  # builds the Clay library
lake env lean AxiomReport.lean | grep AXSUMMARY
#   expected at 43653aa: AXSUMMARY 125 theorems, 236 declarations; nonstandard: 0
#   expected after the smoothness-index correction: 133 theorems, 244 declarations
lake env lean SolutionLinkedRegression.lean
lake env lean ../papers/spatial-bridge/PaperChecks.lean   # paper branch only
cd ../papers/spatial-bridge
python3 check_refs.py       # every file:line in the paper names its declaration
tectonic spatial_bridge.tex # builds spatial_bridge.pdf
```

In CI the clean library build takes about four minutes with the Mathlib cache.

**Acknowledgement of method.** The Lean development and this paper were prepared with AI-assisted tooling under the author's direction. The author is responsible for the content. All formal claims are those checked by the Lean kernel as described above.

## References

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[3] Jian Deng, Thomas Y. Hou, and Xinwei Yu. Geometric properties and nonblowup of 3D incompressible Euler flow. *Communications in Partial Differential Equations*, 30:225–243, 2005. Preprint: https://arxiv.org/abs/math-ph/0402032. Section and equation numbers cited in this paper refer to the arXiv version.

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[5] Benjamin Stanley Frohman. BenFrohman/Clay-: Lean 4 formalization of Fefferman statement (B) and boundary ledger. https://github.com/BenFrohman/Clay-, 2026. Commit 43653aab3b51269a71a713ccbc4d892869bbe497.

[6] Benjamin Stanley Frohman. Fefferman statement (B) on $\mathbb{T}^3$: A precise Lean formalization, a kernel-checked reduction, and the boundaries of the Frohmanian symplectic tether. Repository file paper/Clay_Technical_Monograph_2026.tex, https://github.com/BenFrohman/Clay-, 2026. Concept DOI 10.5281/zenodo.23653012.

[7] The mathlib Community. The Lean mathematical library. In *Proceedings of the 9th ACM SIGPLAN International Conference on Certified Programs and Proofs (CPP 2020)*, pages 367–381. ACM, 2020.